

Formation Engine Methodology

Seed & Bead Model

6+9 Unit Steady-State Step Function

Unified Atomistic–Coarse-Grained Integration with Environment-Responsive Dynamics

VSEPR-SIM Project

Version 3.0.0 — First Implementation Draft

Abstract

This document specifies the *Seed & Bead Model*, a unified simulation step function that combines six atomistic (“seed”) units with nine coarse-grained (“bead”) units into a single deterministic steady-state integrator. The seed layer provides geometry prediction, Velocity Verlet propagation, FIRE quenching, force evaluation, statistical accumulation, and convergence detection. The bead layer provides CG mapping, three fast environment observables (ρ_B , C_B , $P_{2,B}$), a slow internal state (η_B) with relaxation dynamics, target function evaluation, and three-channel kernel modulation (γ_{steric} , γ_{elec} , γ_{disp}). The composite step function achieves steady state when both the atomistic force convergence and the environment-state convergence criteria are simultaneously satisfied. Every sub-unit produces an inspectable diagnostic per step. Implementation: `coarse_grain/models/seed_bead_stepper.hpp`.

Contents

§1 Introduction

The formation engine has historically operated as a sequential pipeline:

$$\text{VSEPR} \rightarrow \text{MD} \rightarrow \text{FIRE} \rightarrow \text{CG mapping} \rightarrow \text{Classification}$$

This is adequate for small molecules but insufficient for environment-responsive bead dynamics, where the coarse-grained layer must feed information back into the force evaluation at every step.

The Seed & Bead model replaces the sequential pipeline with a *composite step function* that executes all 15 sub-units within a single timestep. The two layers operate concurrently:

Layer	Units	Purpose	Convergence
SEED	S1–S6	Atomistic structure	$ F_{\text{rms}} < F_{\text{tol}}$
BEAD	B1–B9	Environment response	$ \Delta\eta _{\text{max}} < \eta_{\text{tol}}$

Steady state is defined as simultaneous convergence of both layers.

1.1 Design Principles

The model follows the project’s anti-black-box philosophy:

1. Every sub-unit produces an inspectable diagnostic per step.
2. Identical inputs produce bit-identical outputs (deterministic).
3. Every parameter is explicitly declared and bounded.
4. No hidden state: all variables are accessible to the reporter.

§2 SEED Layer — 6 Atomistic Units

The SEED layer operates on the bead system as if it were an atomistic system: positions, velocities, forces, and energies. The six units execute in order S4, S1, S3, S2, S5, S6 (force evaluation precedes integration to support FIRE mixing).

2.1 S1: VSEPR Geometry Prediction

The initial geometry is established by the VSEPR domain model, which assigns ideal bond angles based on electron-domain count. During steady-state integration, S1 acts as a classification anchor: the geometry class (linear, trigonal planar, tetrahedral, etc.) is recorded in the step diagnostic but does not modify forces.

2.2 S2: Velocity Verlet Integration

Position and velocity are propagated via the standard Velocity Verlet scheme in real units (Å, amu, fs, kcal/mol):

$$\mathbf{v}_i\left(t + \frac{\Delta t}{2}\right) = \mathbf{v}_i(t) + \frac{\Delta t}{2} \frac{\mathbf{F}_i(t)}{m_i} \quad (1)$$

$$\mathbf{r}_i(t + \Delta t) = \mathbf{r}_i(t) + \Delta t \mathbf{v}_i\left(t + \frac{\Delta t}{2}\right) \quad (2)$$

The timestep Δt is controlled by the FIRE algorithm (S3) and ranges from $\Delta t_{\min}$ to $\Delta t_{\max}$.

2.3 S3: FIRE Quench

The Fast Inertial Relaxation Engine (FIRE) adaptively mixes velocity with the force direction:

$$\mathbf{v}_i \leftarrow (1 - \alpha) \mathbf{v}_i + \alpha |\mathbf{v}| \hat{\mathbf{F}}_i \quad (3)$$

The power $P = \mathbf{F} \cdot \mathbf{v}$ controls adaptation:

Condition	Action
$P > 0$ for $N_{\min}$ consecutive steps	$\Delta t \rightarrow \min(\Delta t \cdot f_{\text{inc}}, \Delta t_{\max}), \alpha \rightarrow \alpha \cdot f_{\alpha}$
$P \leq 0$	$\Delta t \rightarrow \Delta t \cdot f_{\text{dec}}, \alpha \rightarrow \alpha_0, \mathbf{v} \rightarrow \mathbf{0}$

Default parameters: $\alpha_0 = 0.1$, $N_{\min} = 5$, $f_{\text{inc}} = 1.1$, $f_{\text{dec}} = 0.5$, $f_{\alpha} = 0.99$.

2.4 S4: Force Evaluation

Pairwise bead forces are computed using the LJ 12-6 potential with Lorentz–Berthelot combining rules and environment modulation:

$$U_{ij}^{\text{mod}} = 4\varepsilon_{ij} g_{\text{disp}}(\eta_i, \eta_j) \left[\left(\frac{\sigma_{ij}}{r_{ij}} \right)^{12} - \left(\frac{\sigma_{ij}}{r_{ij}} \right)^6 \right] \quad (4)$$

where g_{disp} is the dispersion-channel modulation factor from the BEAD layer (B9). The force is the negative gradient:

$$\mathbf{F}_i = -\nabla_{\mathbf{r}_i} \sum_{j>i} U_{ij}^{\text{mod}} \quad (5)$$

Combining rules:

$$\sigma_{ij} = \frac{1}{2}(\sigma_i + \sigma_j), \quad \varepsilon_{ij} = \sqrt{\varepsilon_i \varepsilon_j} \quad (6)$$

2.5 S5: Statistical Accumulation

Online mean and variance of the total energy are accumulated via Welford’s algorithm:

$$\bar{E}_n = \bar{E}_{n-1} + \frac{E_n - \bar{E}_{n-1}}{n} \quad (7)$$

$$M_{2,n} = M_{2,n-1} + (E_n - \bar{E}_{n-1})(E_n - \bar{E}_n) \quad (8)$$

The running variance $\sigma_E^2 = M_{2,n}/(n-1)$ is used for convergence monitoring.

2.6 S6: Convergence Detection

The SEED layer is considered converged when:

$$F_{\text{rms}} = \sqrt{\frac{1}{N} \sum_{i=1}^N |\mathbf{F}_i|^2} < F_{\text{tol}} \quad (9)$$

Default: $F_{\text{tol}} = 0.05 \text{ kcal/mol/\AA}$.

§3 BEAD Layer — 9 Environment-Responsive Units

The BEAD layer extends each bead with the environment-responsive state $X_B = \{\rho_B, C_B, P_{2,B}, \eta_B\}$ and couples it back into the interaction kernels. The nine units execute in sequence B1–B9.

3.1 B1: CG Mapping

The CG mapping is implicit: bead positions are the primary state variables. B1 confirms that the mapping is up to date and sets the diagnostic flag. In future multi-resolution extensions, B1 will trigger re-mapping from an underlying atomistic state.

3.2 B2: Local Density ρ_B

$$\rho_B = \sum_{C \neq B} w(r_{BC}), \quad w(r) = e^{-r^2/2\sigma_\rho^2} \cdot f_{\text{sw}}(r; r_c, \delta) \quad (10)$$

Normalised: $\hat{\rho}_B = \min(\rho_B/\rho_{\text{max}}, 1)$.

Default parameters: $\sigma_\rho = 3.0 \text{ \AA}$, $r_c = 8.0 \text{ \AA}$, $\delta = 1.0 \text{ \AA}$, $\rho_{\text{max}} = 10$.

3.3 B3: Coordination Number C_B

$$C_B = \sum_{C \neq B} \mathbf{1}_{[r_{BC} < r_c]} \quad (11)$$

This is the hard-cutoff count of neighbours within r_c .

3.4 B4: Orientational Order $P_{2,B}$

$$P_{2,B} = \frac{1}{|\mathcal{N}_B|} \sum_{C \in \mathcal{N}_B} \frac{1}{2} (3 \cos^2 \theta_{BC} - 1) \quad (12)$$

where $\cos \theta_{BC} = \hat{n}_B \cdot \hat{n}_C$ and $\hat{n}_B$ is the primary orientation axis of bead B .

Normalised: $\hat{P}_{2,B} = (P_{2,B} + 0.5)/1.5 \in [0, 1]$.

3.5 B5: Target Function

The target function combines normalised fast observables:

$$f_B = \alpha \hat{\rho}_B + \beta \hat{P}_{2,B} \quad (13)$$

with $\alpha + \beta = 1$, $\alpha, \beta \in [0, 1]$. Default: $\alpha = 0.5$, $\beta = 0.5$.

3.6 B6: Slow State η_B Integration

The slow state η_B evolves via first-order relaxation with the exact exponential solution (unconditionally stable):

$$\eta_B(t + \Delta t) = f_B + (\eta_B(t) - f_B) e^{-\Delta t/\tau} \quad (14)$$

Steady-state value: $\eta_B^{\text{ss}} = f_B(\rho, C, P_2)$.

The relaxation timescale τ controls how quickly the bead's internal state responds to environmental changes. Large τ produces hysteresis; small τ produces instantaneous response.

Default: $\tau = 100$ fs.

3.7 B7–B9: Kernel Modulation

The environment state modulates interaction kernels through the Option A coupling (recommended by the ERB specification):

$$K_k(\ell, r; \eta_A, \eta_B) = K_k(\ell, r) \cdot (1 + \gamma_k \bar{\eta}_{AB}) \quad (15)$$

where $\bar{\eta}_{AB} = \frac{1}{2}(\eta_A + \eta_B)$ and k indexes the channel:

Unit	Channel	γ_k	Physical effect
B7	Steric	$\gamma_{\text{steric}} \in [0, 1]$	Hardening under compression
B8	Electrostatic	$\gamma_{\text{elec}} \in [-1, 0]$	Screening under crowding
B9	Dispersion	$\gamma_{\text{disp}} \in [0, 2]$	Enhancement under ordering

Invariant: $|1 + \gamma_k \bar{\eta}| > 0$ for all channels (kernel must not change sign).

§4 Complete 6+9 Parameter Set

The Seed & Bead model introduces 15 active parameters, partitioned across the two layers:

Layer	Symbol	Name	Default	Unit
S	Δt_0	Initial timestep	1.0	fs
S	$\Delta t_{\max}$	Maximum FIRE timestep	10.0	fs
S	F_{tol}	Force convergence	0.05	kcal/mol/Å
S	E_{tol}	Energy convergence (relative)	10^{-6}	—
S	α_0	FIRE mixing parameter	0.1	—
S	$N_{\max}$	Maximum step count	10000	—
B	α	Density weight	0.5	—
B	β	Orientation weight	0.5	—
B	τ	Relaxation timescale	100	fs
B	γ_{steric}	Steric modulation	0.2	—
B	γ_{elec}	Electrostatic modulation	-0.1	—
B	γ_{disp}	Dispersion modulation	0.5	—
B	σ_ρ	Density Gaussian width	3.0	Å
B	r_c	Observable cutoff	8.0	Å
B	η_{tol}	η convergence	10^{-3}	—

The “6+9” designation refers to 6 SEED units and 9 BEAD units, not the parameter count.

§5 Steady-State Condition

The composite system has reached steady state when all 15 units have converged. In practice, this reduces to two independent criteria:

$$\boxed{\text{Steady state} \iff F_{\text{rms}} < F_{\text{tol}} \wedge \max_B |\Delta\eta_B| < \eta_{\text{tol}}} \quad (16)$$

The first condition ensures the mechanical structure is relaxed. The second condition ensures the environment-responsive state has equilibrated. Both must be satisfied simultaneously.

5.1 Convergence Hierarchy

The two criteria converge on different timescales:

- SEED convergence is typically faster ($\sim 10^2$ steps for small systems with FIRE acceleration).
- BEAD convergence depends on τ : the relaxation time determines how many steps are needed for η to reach f within tolerance.

For $\Delta t/\tau \ll 1$, the number of steps to converge η from η_0 to f within tolerance η_{tol} is:

$$N_\eta \approx \frac{\tau}{\Delta t} \ln\left(\frac{|f - \eta_0|}{\eta_{\text{tol}}}\right) \quad (17)$$

§6 Step Execution Order

Within a single call to `SeedBeadStepper::step()`, the 15 units execute in the following order:

Order	Unit	Action	Depends on
1	S4	Evaluate forces	positions, η (from previous step)
2	S1	Record geometry	positions
3	S3	FIRE velocity mix	forces, velocities
4	S2	Verlet integration	forces, velocities, Δt
5	S5	Accumulate energy	positions, η
6	S6	Check convergence	forces
7	B1	CG mapping check	positions
8	B2	Compute ρ_B	neighbour distances
9	B3	Compute C_B	neighbour count
10	B4	Compute $P_{2,B}$	neighbour orientations
11	B5	Compute f_B	$\hat{\rho}_B, \hat{P}_{2,B}$
12	B6	Integrate η_B	f_B , previous η_B , Δt , τ
13	B7	Modulate steric	$\eta_A, \eta_B, \gamma_{\text{steric}}$
14	B8	Modulate elec.	$\eta_A, \eta_B, \gamma_{\text{elec}}$
15	B9	Modulate disp.	$\eta_A, \eta_B, \gamma_{\text{disp}}$

The modulation factors (B7–B9) computed at step n are consumed by S4 at step $n + 1$. This one-step lag is the standard approach for explicit integration and introduces no instability when $\Delta t/\tau < 0.5$.

§7 Step Diagnostic Record

Every call to `step()` returns a `SeedBeadStepRecord` containing the full diagnostic state. No information is hidden.

7.1 SEED Diagnostics

Field	Type	Description
<code>total_energy</code>	<code>double</code>	Total system energy (kcal/mol)
<code>rms_force</code>	<code>double</code>	F_{rms} (kcal/mol/Å)
<code>max_force</code>	<code>double</code>	Maximum single-particle force
<code>kinetic_energy</code>	<code>double</code>	$\sum \frac{1}{2} m_i v_i^2$
<code>potential_energy</code>	<code>double</code>	Sum of pairwise U_{ij}^{mod}
<code>dt_current</code>	<code>double</code>	FIRE-adjusted timestep (fs)
<code>seed_status</code>	<code>SeedUnitStatus</code>	Per-unit boolean flags

7.2 BEAD Diagnostics

Field	Type	Description
<code>avg_rho</code>	<code>double</code>	System-averaged ρ
<code>avg_C</code>	<code>double</code>	System-averaged C
<code>avg_P2</code>	<code>double</code>	System-averaged P_2
<code>avg_eta</code>	<code>double</code>	System-averaged η
<code>avg_target_f</code>	<code>double</code>	System-averaged f
<code>max_delta_eta</code>	<code>double</code>	$\max_B \Delta\eta_B $ this step
<code>avg_g_steric</code>	<code>double</code>	Mean steric modulation factor
<code>avg_g_elec</code>	<code>double</code>	Mean electrostatic modulation factor
<code>avg_g_disp</code>	<code>double</code>	Mean dispersion modulation factor
<code>bead_status</code>	<code>BeadUnitStatus</code>	Per-unit boolean flags

§8 Snapshot & Graph Infrastructure

The `SnapshotGraphCollector` class accumulates step diagnostics for automatic report generation. It maintains 14 time-series channels and frame-level position snapshots at configurable intervals.

8.1 Time-Series Channels

Channel	Unit	Source
Energy	kcal/mol	S5
RMS Force	kcal/mol/Å	S6
Max Force	kcal/mol/Å	S4
Mean η	—	B6
Mean ρ	—	B2
Mean C	—	B3
Mean P_2	—	B4
Mean f	—	B5
$\max \Delta\eta $	—	B6
Δt	fs	S3
g_{steric}	—	B7
g_{elec}	—	B8
g_{disp}	—	B9
Kinetic energy	kcal/mol	S2

8.2 Frame Snapshots

At intervals of `snapshot_interval` steps (default: 10) and at steady state, the collector stores a `BeadFrameSnapshot` containing:

- Bead positions ($\mathbf{r}_i$)
- Per-bead η_i , ρ_i
- Total energy, RMS force
- Steady-state flag

These snapshots feed the XY/XZ trajectory projections in the Python visualization pipeline (`scripts/visualize_seed_bead.py`).

§9 Metal & Gas Study Framework

The `metal_gas_study.hpp` module automates the generation of material-specific study reports using the experimental reference data from `atomistic/validation/nanoparticle_ref.hpp`.

9.1 Reference Materials

Label	Z	Class	a_0 (Å)	CN	Source
Au 5nm FCC	79	Metal (FCC)	4.078	12.0	Cleveland1997
Ag 5nm FCC	47	Metal (FCC)	4.086	12.0	Tyson1977
Cu 4nm FCC	29	Metal (FCC)	3.615	12.0	CRC105
Ar 5nm	18	Noble Gas	5.260	12.0	Baletto2005
Pt 5nm FCC	78	Metal (FCC)	3.924	12.0	Tyson1977
Fe 5nm BCC	26	Metal (BCC)	2.870	8.0	CRC105

9.2 Study Workflow

For each material:

1. Build a `BeadSystem` from the reference lattice constant and estimated LJ parameters.
2. Configure `SeedBeadParams` based on material class (FCC metal / BCC metal / noble gas).
3. Run `SeedBeadStepper` to steady state.
4. Compare final observables against the `ExperimentalRecord`.
5. Generate a Markdown report with per-metric pass/fail verdicts.
6. Export time-series and snapshot CSV files.

9.3 Comparison Metrics

Each study produces the following comparison metrics:

Metric	Simulated	vs	Experimental
Coordination C	$\langle C \rangle_{\text{final}}$	$< \text{tol}$	Bulk CN
Cohesive energy	E/N	$< \text{tol}$	E_{coh} per atom
Steady-state η	$\langle \eta \rangle_{\text{final}}$	—	(informational)

§10 Live Visualization

The `SeedBeadViewer` class provides a real-time 60 fps GLFW/ImGui viewer for the seed-and-bead step function. It is the primary visual interface for inspecting environment-responsive dynamics.

10.1 Render Loop Architecture

1. **Input:** Mouse orbit/pan/zoom, keyboard controls.
2. **Simulation:** Execute `steps_per_frame` calls to `SeedBeadStepper::step()`.
3. **Render:** Draw beads as spheres with overlay color, draw bonds as lines.
4. **UI:** ImGui panels for control, energy graph, force graph, η convergence, bead inspector.
5. **Swap:** `glfwSwapBuffers` with vsync (= 60 fps).

10.2 Overlay Modes

Bead color is mapped from a scalar overlay value through a jet colormap:

Mode	Scalar
η	Slow state $\eta_B \in [0, 1]$
ρ	Normalised density $\hat{\rho}_B$
C	$C_B/12$ (clamped)
P_2	Normalised $\hat{P}_{2,B}$
f	Target function f_B
g_{steric}	Steric modulation factor
g_{elec}	Electrostatic modulation factor
g_{disp}	Dispersion modulation factor
Energy	Proxy: $0.5\hat{\rho} + 0.5\eta$
Uniform	Constant 0.5 (no overlay)

§11 Data Export Formats

The Seed & Bead system exports simulation data in six formats, covering research analysis (CSV, Excel), engineering CAD integration (SolidWorks), and standard point-cloud exchange (XYZ).

11.1 CSV Export

Two CSV files are produced by `SnapshotGraphCollector`:

File	Content	Columns	Rows
Timeseries	Step-level diagnostics	15	1 per step
Snapshots	Per-bead XYZ + η , ρ	7	1 per bead per snapshot

11.2 Excel XML Export

The `export_excel_xml()` function generates a SpreadsheetML 2003 XML file that opens directly in Microsoft Excel, LibreOffice Calc, and Google Sheets. Three worksheets are produced:

1. **Timeseries** — all 15 diagnostic channels with styled headers and unit rows.
2. **Snapshots** — per-bead positions (x , y , z), η , and ρ at each snapshot interval.
3. **Summary** — final-value table for key metrics.

The Python pipeline (`visualize_seed_bead.py -excel`) can additionally produce `.xlsx` format via `openpyxl`, with auto-width columns, number formatting, and colour-coded headers.

11.3 SolidWorks Export

Three SolidWorks-compatible formats are exported by `solidworks_export.hpp`:

Extension	Format	SolidWorks Import Path
<code>.sldcrv</code>	Tab-delimited $X Y Z$	Insert → Curve → Curve Through XYZ Points
<code>.xyz</code>	Standard XYZ point cloud	File → Open (All Files)
<code>.csv</code>	X, Y, Z, η, ρ metadata	Macro or plugin import

The `.sldcrv` format is the native SolidWorks curve input. Per-bead trajectory curves are also exported as individual `.sldcrv` files (`bead_000.sldcrv`, etc.), allowing SolidWorks to reconstruct full bead paths as 3D spline curves.

11.4 Export Pipeline

```
// C++ --- from SnapshotGraphCollector
collector.export_timeseries_csv("data/timeseries.csv");
collector.export_snapshots_csv("data/snapshots.csv");
collector.export_excel("data/report.xml", "Au_5nm_Study");
collector.export_solidworks_curve("data/structure.sldcrv");
collector.export_solidworks_pointcloud("data/structure.xyz");
collector.export_solidworks_metadata_csv("data/points.csv");

# Python --- from CLI
python visualize_seed_bead.py --demo --excel --solidworks
```

§12 Implementation Summary

Component	File	Lines
6+9 Step Function	coarse_grain/models/seed_bead_stepper.hpp	~430
Snapshot & Graph	coarse_grain/report/snapshot_graph.hpp	~400
Metal & Gas Studies	coarse_grain/report/metal_gas_study.hpp	~340
Excel XML Export	coarse_grain/report/excel_export.hpp	~210
SolidWorks Export	coarse_grain/report/solidworks_export.hpp	~200
Live 60 fps Viewer	coarse_grain/vis/seed_bead_viewer.hpp	~410
Demo Application	apps/seed_bead_demo.cpp	~100
Python Visualization	scripts/visualize_seed_bead.py	~650
Test Suite	tests/test_seed_bead_stepper.cpp	~220
LaTeX Document	docs/section_seed_bead_model.tex	(this file)

12.1 Build Targets

```
# Test suite
add_executable(test_seed_bead_stepper test_seed_bead_stepper.cpp)
target_link_libraries(test_seed_bead_stepper coarse_grain atomistic vsepr_core)

# Live demo (with BUILD_VIS=ON)
add_executable(seed-bead-demo apps/seed_bead_demo.cpp)
target_link_libraries(seed-bead-demo coarse_grain atomistic vsepr_vis ...)

# Python visualization
python scripts/visualize_seed_bead.py --demo
```

12.2 Dependencies

Depends on	Provides
environment_state.hpp	X_B , fast observables, η integration
environment_coupling.hpp	Kernel modulation (Option A)
interaction_engine.hpp	Per-channel pairwise energy
bead.hpp, bead_system.hpp	Bead data structures
nanoparticle_ref.hpp	Experimental reference data
GLFW, GLEW, Dear ImGui	Visualization (optional)
matplotlib, numpy	Python figures (optional)
openpyxl	Python Excel .xlsx export (optional)

§13 Relation to Existing Theory Documents

The Seed & Bead model unifies concepts from several existing sections:

Document	Contribution to this model
Section 5 (Integration)	Velocity Verlet, FIRE algorithm
Section 6 (Formation Physics)	VSEPR $\rightarrow$ MD $\rightarrow$ FIRE pipeline
Section 7 (Statistical Interp.)	Welford accumulator, convergence
ASB (Anisotropic Beads)	SH descriptors, channel kernels
ERB (Environment-Responsive)	X_B , η dynamics, kernel modulation
CGL (Coarse-Grained Layer)	CG mapping, ensemble proxies
TIF (Testing Infrastructure)	Validation methodology

The Seed & Bead model is the *first implementation* that executes all these components within a single step function, rather than as a sequential pipeline.

13.1 Terminology Compliance

Per project rules (`.github/copilot-instructions.md`, §6):

- The term “atomistic” is used throughout.
- The forbidden term is not present in any code, comment, or documentation artifact produced by this model.

End of Seed & Bead Model specification.

VSEPR-SIM v2.8.0 — Deterministic atomistic platform.